

E-Commerce Sales & Customer Analytics System

Business Insights & Analysis Report

This report summarizes SQL-based business analysis of the cleaned Superstore e-commerce dataset, covering sales, profit, products, customers, regions, discounts, and yearly trends.

1. Executive Summary

The dataset contains 9,994 records covering orders from January 2014 to December 2017. The business generated \$2.30 million in sales and \$286.40 thousand in profit across 5,009 orders and 793 customers, with an overall profit margin of 12.47%. The analysis identifies substantial differences in profitability across categories, regions, products, customers, and discount levels.

2. Key Performance Indicators

KPI	Result
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Orders	5,009
Total Customers	793
Average Order Value	\$458.61
Profit Margin	12.47%

3. Category Analysis

Category	Sales	Profit	Margin
Technology	\$836,154.10	\$145,455.70	17.39%
Furniture	\$742,000.00	\$18,451.25	2.49%
Office Supplies	\$719,047.00	\$122,490.90	17.03%

Finding: Technology has the highest sales (\$836.15K) and profit (\$145.46K) among the three categories, with a 17.39% margin. Furniture generates \$742.00K in sales but only \$18.45K in profit, resulting in a much lower 2.49% margin.

4. Regional Analysis

Region	Sales	Profit	Margin
Central	\$501,239.89	\$39,706.36	8%
East	\$678,781.24	\$91,522.78	13%
South	\$391,721.91	\$46,749.43	12%
West	\$725,457.82	\$108,418.45	15%

Finding: West records \$725.46K in sales and \$108.42K in profit, while Central records \$501.24K in sales and \$39.71K in profit. Central has the lowest regional profit margin at 8%, compared with 15% in West.

5. Product Analysis

High-sales products: Canon imageCLASS 2200 Advanced Copier generated \$61,599.82 in sales and \$25,199.93 in profit. However, some products with substantial sales were loss-making, including Cisco TelePresence System EX90 (\$22,638.48 sales; -\$1,811.08 profit) and GBC DocuBind P400 (\$17,965.07 sales; -\$1,878.17 profit).

Largest observed product losses: Cubify CubeX 3D Printer Double Head Print: -\$8,879.97; Lexmark MX611dhe Monochrome Laser Printer: -\$4,589.97; Cubify CubeX 3D Printer Triple Head Print: -\$3,839.99; Chromcraft Bull-Nose Wood Oval Conference Tables & Bases: -\$2,876.11; Bush Advantage Collection Racetrack Conference Table: -\$1,934.40.

6. Discount Analysis

Discount	Sales	Profit	Margin
0%	\$1,087,908	\$320,988	30%
10%	\$54,369	\$9,029	17%
15%	\$27,559	\$1,419	5%
20%	\$764,594	\$90,337	12%
30%	\$103,227	-\$10,369	-10%
32%	\$14,493	-\$2,391	-16%
40%	\$116,418	-\$23,057	-20%
45%	\$5,485	-\$2,493	-45%
50%	\$58,919	-\$20,506	-35%
60%	\$6,645	-\$5,945	-89%
70%	\$40,620	-\$40,075	-99%
80%	\$16,964	-\$30,539	-180%

Finding: In the aggregated dataset, discount levels of 30% and above are associated with negative profit. For example, 30% discount has a -10% margin, while 70% and 80% discounts have -99% and -180% margins. This establishes an important profitability signal, but does not by itself prove that discounting causes the losses.

7. Yearly Trend Analysis

Year	Sales	Profit
2014	\$484,247.60	\$49,544.06
2015	\$470,532.50	\$61,618.69
2016	\$609,205.90	\$81,795.27
2017	\$733,215.20	\$93,439.77

Finding: Sales and profit increased from 2015 through 2017. Sales rose from \$470.53K in 2015 to \$733.22K in 2017, while profit rose from \$61.62K to \$93.44K. In 2014–2015, sales decreased slightly while profit increased.

8. Customer Analysis

Finding: Customer revenue does not necessarily correspond to customer profitability. Sean Miller generated \$25,043.07 in sales but recorded a loss of \$1,980.75 and a -7.91% margin. In contrast, Tamara Chand generated \$19,052.22 in sales and \$8,981.32 in profit, a 47.14% margin.

9. Key Business Problems

- Furniture has a substantially lower profit margin (2.49%) than Technology (17.39%) and Office Supplies (17.03%).
- Several individual products generate significant losses despite generating sales.
- Central has the lowest regional profit margin at 8%.
- Aggregated transactions with discount levels of 30% or higher show negative profit.
- Some high-sales customers can still be loss-making, as demonstrated by Sean Miller.

10. Business Opportunities

- Investigate Furniture product, pricing, cost, and discount patterns.
- Review major loss-making products and their pricing, discounts, and costs.
- Investigate Central-region product mix, discounts, and customer patterns.

- Review high-discount transactions and consider stronger discount controls where profitability is negative.
- Analyze profitable customer and product patterns and identify successful combinations.

11. Recommendations

- Conduct a detailed profitability review of Furniture, focusing on sub-categories and individual products.
- Create a loss-making product watchlist and review pricing, discount, and cost structure.
- Perform a Central-region diagnostic using product mix, discount, and customer-level analysis.
- Use data-driven discount thresholds or approval rules for high-discount transactions while monitoring sales and profit.
- Evaluate customers using profitability as well as sales when assessing commercial value.

12. Conclusion

The analysis shows that sales volume alone is not sufficient to understand business performance. Category, region, product, customer, and discount-level profitability vary considerably. The main areas for management attention are Furniture profitability, loss-making products, Central-region margins, high-discount transactions, and customer-level profitability. These areas can be explored further in the Excel, Python, and Power BI stages.

Note: Values and percentages are based on the analysis results provided during the project workflow. Where the current analysis shows association rather than causation, the report avoids presenting a causal conclusion.